

Pre-processing pipeline

- 1- Co-register all modalities.
- 2- Do skull stripping – you can use ANTs forward transformation script for that.
- 3- Perform bias correction on (2).
- 4- Obtain the tumor mask using either nnUnet model or manual delineation
- 5- Perform ANTs diffeomorphic transformation to register the subject in (3) to the MNI space (skull-stripped and bias corrected); this assures forward and inverse transforms are applied only once
- 6- Perform intensity normalization on Step 5 result.

Your output file that you are using for the deformation pipeline should look like this:

```
'T1_SS_BiasCorrect_MNI_InverseWarp_DeformMag_Normalized.nii.gz'
```

Additionally, prior to extracting your deformation value statistics, you may need to apply linear scaling or Z-score normalization to make sure all values are in the same range from your deformation magnitude file output by ANTs.

Example:

```
deform=load_untouch_nii('T1_SS_BiasCorrect_MNI_InverseWarp_DeformMag_Normalized.nii.gz');
```

```
%%Apply linear scaling
```

```
deform.img = (deform.img - min(min(min(deform.img))))./(max(max(max(deform.img)))-  
min(min(min(deform.img))));
```